# Hints: Methods of proof – 4. Invariants, monovariants and double counting

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

---

## Problem 1

**Hint 1.** Treat the empty cell as a sixteenth tile, so a position is a permutation $\pi$ of the 16 cells. Every move is a transposition of the empty cell with a neighbouring tile. What does a move do to $\operatorname{sgn}\pi$, and what does it do to the colour of the empty cell in the chessboard colouring?

**Hint 2.** Each move changes both: the sign flips and the empty cell changes colour. So $I = \operatorname{sgn}\pi \cdot (-1)^c$, where $c$ is the colour (equivalently, the parity of the taxicab distance of the empty cell from its home), is invariant.

**Hint 3.** Swapping two tiles with the empty cell at home changes $\operatorname{sgn}\pi$ but not $c$, so it changes $I$: unreachable. General principle: the reachable configurations from $x$ form the orbit of $x$ under the group (or semigroup) generated by the moves; any function constant on these orbits, typically $x \mapsto \chi(g)$ for a homomorphism $\chi$ (here to $\{\pm 1\}$) that is trivial on all moves, separates orbits.

## Problem 2

**Hint 1.** The map $(a, b) \mapsto \left(\frac{a+b}{\sqrt{2}}, \frac{a-b}{\sqrt{2}}\right)$ is given by an orthogonal $2 \times 2$ matrix. Which simple symmetric functional of the numbers on the board does it preserve?

**Hint 2.** The sum of squares is invariant, so a multiset with a different sum of squares is unreachable. It is not sufficient. With two numbers write the board as a point $(a, b)$ of the plane; the move is the reflection in the line at angle $22.5°$, and since the order of the two numbers is free, the swap (reflection in the line at $45°$) is available too. Their product is the rotation by $45°$, so they generate the symmetry group of a regular octagon (order 16), and the reachable multisets are exactly $\{\pm a, \pm b\}$ and $\left\{\pm\frac{a+b}{\sqrt{2}}, \pm\frac{a-b}{\sqrt{2}}\right\}$ with all choices of signs: at most 8 multisets on the circle $x^2 + y^2 = a^2 + b^2$, although the invariant allows the whole circle.

**Hint 3.** For $(a, b) \mapsto (a + b, ab)$: the new pair is the root pair of $(x - s)(x - p)$, where $s = a + b$, $p = ab$, so on $(s, p)$ the operation acts as $(s, p) \mapsto (s+p, sp)$; no symmetric polynomial is preserved. Use a monovariant instead: the sum grows by $ab$, so on positive numbers the sum strictly increases, hence no configuration occurs twice, and only multisets with larger sum (and positive entries) can be reached. (Positivity matters: a 0 on the board is frozen, since $(a, 0) \mapsto (a, 0)$, and a negative product decreases the sum.)

## Problem 3

**Hint 1.** For positive integers $a \geqslant b > 0$ the sum $a + b$ (or the larger number) strictly decreases at each step and is a positive integer, so the process stops, namely when one number becomes 0.

**Hint 2.** The invariant is $\gcd(a, b) = \gcd(a - b, b)$; at the end it is the remaining non-zero number. Grouping consecutive subtractions of the same $b$ gives division with remainder: the Euclidean algorithm.

**Hint 3.** Complexity: subtraction alone can take $a$ steps (e.g. $(a, 1)$). With division, if $r_k = qr_{k+1} + r_{k+2}$, then $r_k \geqslant r_{k+1} + r_{k+2}$, so the remainders dominate Fibonacci numbers backwards; the number of division steps is $O(\log b)$ (Lamé: at most about $\log_\varphi b + 1$; also $r_{k+2} < r_k/2$).

**Hint 4.** One continuous analogue: for the gradient flow $x' = -\nabla V(x)$ the energy $V$ is a Lyapunov function, $V(x(t)) = V(x(0)) - \int_0^t |x'(s)|^2\,ds$. If $V$ is bounded below, $\int_0^\infty |x'|^2 < \infty$, which is the continuous form of "a bounded monovariant cannot decrease forever". (Note also that Euclid on reals $a, b$ terminates iff $a/b \in \mathbb{Q}$.)

## Problem 4

**Hint 1.** $\sum_v \deg v = 2|E|$: count pairs $(v, e)$ with $v$ an endpoint of $e$. For the 4-cycle bound count the "cherries" $(v, \{x, y\})$ with $x \neq y$ both adjacent to $v$.

**Hint 2.** By vertices $v$ there are $\sum_v \binom{d_v}{2}$ cherries; by pairs $\{x, y\}$ there are at most $\binom{n}{2}$, because without a $C_4$ two vertices have at most one common neighbour. Hence $\sum_v \binom{d_v}{2} \leqslant \binom{n}{2}$.

**Hint 3.** By convexity (Jensen), $n\binom{d}{2} \leqslant \binom{n}{2}$ with $d = 2|E|/n$, i.e. $d^2 - d - (n-1) \leqslant 0$, so $d \leqslant \frac{1+\sqrt{4n-3}}{2}$ and $|E| = \frac{nd}{2} \leqslant \frac{1}{2}\big(1 + \sqrt{4n-3}\big) \cdot \frac{n}{2}$.

**Hint 4.** Oddtown: sets $A_1, \ldots, A_m \subset \{1, \ldots, n\}$ with odd sizes and even pairwise intersections. Their characteristic vectors over $\mathbb{F}_2$ have Gram matrix $I_m$, which has rank $m$, so the vectors are independent and $m \leqslant n$. Comparison: double counting bounds a quantity by averaging over incidences (and the $C_4$ bound is asymptotically sharp); for Oddtown a counting argument gives only weaker bounds, while the rank gives $m \leqslant n$ at once, which is sharp (take singletons).

## Problem 5

**Hint 1.** Encode $S$ by a generating function: $|S|$ is the coefficient of $x^{3860}$ in $f(x)^{2018}$, where $f(x) = x + x^2 + x^3 + x^4 + x^5 + x^6 + x^{10}$.

**Hint 2.** All coefficients are non-negative, so for every $x > 0$ we have $|S|\, x^{3860} \leqslant f(x)^{2018}$, i.e. $|S| \leqslant f(x)^{2018}/x^{3860}$. Choose $x$ cleverly.

**Hint 3.** Take $x = \frac{1}{2}$: $f(\frac{1}{2}) = \frac{63}{64} + \frac{1}{1024} = \frac{1009}{1024} = \frac{2018}{2048}$, which gives exactly $|S| \leqslant 2^{3860}\big(\frac{2018}{2048}\big)^{2018}$.